

Triangle–Gaussian convolution preserves extremality

Full answer to the first clause of Question 3 in arXiv:0801.0941
Routed from the open problem in arXiv:1803.06409

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Status. The first clause of Question 3 has an affirmative answer:

$$f(x) = (1 - |x|)_+, \quad \gamma(x) = e^{-\pi x^2} \implies f * \gamma \text{ is extremal in } \Omega_1^+.$$

Together with this run's existing counterexample to the second clause, this completes the explicit two-part question.

Source route and exact question

The queue source states that describing the extreme rays of the cone of nonnegative positive-definite $C_0(\mathbb{R})$ functions is open and directs the reader to Jaming–Matolcsi–Révész [1, p. 9].

$$\mathcal{D}_c = \mathcal{D} \cap \mathcal{C}_c(G, \mathbb{R}), \quad \mathcal{D}_0 = \mathcal{D} \cap \mathcal{C}_0(G, \mathbb{R});$$

e) $\mathcal{D}^*$: the set of real valued integrally positive definite functions.

A convex cone is subject to the appropriate version of the Krein–Milman or Choquet theorem, so its closure consists of limits of linear combinations of its extreme points. However, describing the extreme points of e.g. $\mathcal{P}_0 \cap \mathcal{D}_0$ is still an open problem even for the most immediate case of $G = \mathbb{R}$. For more on this problem, attributed to Choquet, the interested reader can consult with the publication [17].

In the lights of the above, one may expect that the dual cone of $\mathcal{P}_0 \cap \mathcal{D}_0$ is even less easy to describe. Still, we will find a straightforward description. Our major tool to the solution of this problem will be an intersection formula on the dual cones of the intersection of two cones.

Routing evidence: arXiv:1803.06409, PDF page 9.

The cited paper defines Ω_1^+ as the cone of continuous nonnegative positive-definite functions on $\mathbb{R}$. It recalls that the triangle

$$f = \mathbf{1}_{[-1/2, 1/2]} * \mathbf{1}_{[-1/2, 1/2]} = (1 - |x|)_+$$

and $\gamma = e^{-\pi x^2}$ are extremal, and asks first whether $f * \gamma$ is extremal [2, p. 18, Question 3].

— Finally, all the extremals we found, except the gaussians, either have at least a zero, or their Fourier transform has at least a zero. Borisov [Bor] conjectured that the only extremals in $L^2(\mathbb{R}) \cap \mathcal{C}(\mathbb{R})$ that are strictly positive and that have strictly positive Fourier transform are the gaussian functions $e^{-a\pi x^2}$, $a > 0$. We are inclined to believe that this is false. This is linked to Question 1 above which leads us to the following more precise question:

Question 3.

Consider the two functions $f = \chi_{[-1/2, 1/2]} * \chi_{[-1/2, 1/2]}$ and $\gamma = e^{-\pi x^2}$, and recall that they are extremal positive positive definite functions in Ω_1^+ . Is $f * \gamma$ extremal in Ω_1^+ and, if so, is $\gamma(f * \gamma)$ extremal in Ω_1^+ ?

A positive answer to the second question would of course provide a counterexample to Borisov's conjecture.

Exact source question: arXiv:0801.0941, PDF page 18.

We use

$$\widehat{a}(\xi) = \int_{\mathbb{R}} a(x) e^{-2\pi i x \xi} dx, \quad F(\xi) := \widehat{f}(\xi) = \left(\frac{\sin \pi \xi}{\pi \xi} \right)^2, \quad \widehat{\gamma} = \gamma.$$

Answer

Theorem 1. *The function $h = f * \gamma$ generates an extremal ray of Ω_1^+ . That is, if $g, h - g \in \Omega_1^+$, then $g = \lambda h$ for some $0 \leq \lambda \leq 1$.*

Proof intuition. An interval component $0 \leq g \leq f * \gamma$ also satisfies the Fourier-side inequality $0 \leq \widehat{g} \leq F\gamma$. Removing the nowhere-zero Gaussian factor defines a bounded “initial datum” u with $g = u * \gamma$. A critical Phragmén–Lindelöf/Paley–Wiener lemma says that domination by the heat evolution of a function supported on $[-1, 1]$ forces u to have the same support bound. Thus $\widehat{u}$ is entire of the same exponential type as F . Its nonnegativity makes it vanish to even order at every nonzero integer, exactly where the sinc square has double zeros. The quotient by F is therefore entire, has polynomial growth, and is bounded on the real axis; it must be constant.

A Gaussian domination support lemma

Lemma 1. *Let u be bounded on $\mathbb{R}$, let $a \geq 0$ be integrable with $\text{supp } a \subseteq [-R, R]$, and suppose*

$$0 \leq (u * \gamma)(x) \leq (a * \gamma)(x) \quad (x \in \mathbb{R}). \quad (1)$$

Then $\text{supp } u \subseteq [-R, R]$.

Proof. Put $v(t) = u(t)e^{-\pi t^2} \in L^1 \cap L^2$, and define

$$L(z) = \int_{\mathbb{R}} v(t)e^{2\pi zt} dt \quad (z \in \mathbb{C}).$$

This is entire and the Gaussian integral gives

$$|L(x + iy)| \leq Ce^{\pi x^2}. \quad (2)$$

Completing the square in (1) gives, for real x ,

$$\begin{aligned} 0 \leq L(x) &= e^{\pi x^2} (u * \gamma)(x) \\ &\leq \int_{-R}^R a(t)e^{-\pi t^2} e^{2\pi xt} dt \leq Ce^{2\pi R|x|}. \end{aligned} \quad (3)$$

Also $|L(iy)| \leq \|v\|_1$.

For completeness, we record the critical-order step that converts these axial bounds into a Paley–Wiener bound. Let

$$\mathfrak{h}(\theta) = \limsup_{r \rightarrow \infty} r^{-2} \log |L(re^{i\theta})|$$

be the order-two indicator. Equation (2) gives $\mathfrak{h}(\theta) \leq \pi \cos^2 \theta$, while the improved bounds give $\mathfrak{h}(0), \mathfrak{h}(\pi/2) \leq 0$. Fix $0 < \theta < \pi/2$ and put $\beta = \pi/2 - \delta > \theta$. The standard two-trigonometric convexity of the indicator on a sector of width less than $\pi/2$ yields

$$\mathfrak{h}(\theta) \leq \frac{\mathfrak{h}(0) \sin 2(\beta - \theta) + \mathfrak{h}(\beta) \sin 2\theta}{\sin 2\beta} \leq \frac{\pi \sin^2 \delta}{\sin 2\delta} \sin 2\theta.$$

Letting $\delta \downarrow 0$ proves $\mathfrak{h} \leq 0$ in that quadrant; the other three are identical. Thus L is of order at most two and minimal type. The critical Phragmén–Lindelöf theorem, applied in the right quadrants to $L(z)e^{-2\pi Rz}$ and in the left quadrants to $L(z)e^{2\pi Rz}$, now gives

$$|L(z)| \leq Ce^{2\pi R|\Re z|}. \quad (4)$$

Indeed, each modified function has minimal type and is bounded on both rays forming the relevant quadrant by (3) and the imaginary-axis bound.

The entire function $W(\zeta) = L(-i\zeta)$ is therefore of exponential type at most $2\pi R$, and $W|_{\mathbb{R}}$ is the Fourier transform of the L^2 -function v . The L^2 Paley–Wiener theorem gives $\text{supp } v \subseteq [-R, R]$. Since $e^{-\pi t^2}$ never vanishes, $\text{supp } u = \text{supp } v$, proving the lemma. $\square$

Proof of Theorem 1

Suppose $g, h - g \in \Omega_1^+$. Then $0 \leq g \leq h$, so both summands are integrable. Their positive definiteness gives

$$0 \leq G(\xi) := \widehat{g}(\xi) \leq \widehat{h}(\xi) = F(\xi)e^{-\pi\xi^2} \quad (\xi \in \mathbb{R}). \quad (5)$$

Define

$$U(\xi) = e^{\pi\xi^2} G(\xi).$$

By (5), $0 \leq U \leq F$. Since $F \in L^1$, also $U \in L^1$. Let $u = \check{U}$; then u is bounded and continuous, and Fourier inversion gives

$$g = u * \gamma.$$

Lemma 1, applied with $a = f$ and $R = 1$, gives $\text{supp } u \subseteq [-1, 1]$. Thus

$$U(z) = \int_{-1}^1 u(t)e^{-2\pi itz} dt$$

extends to an entire function of exponential type at most 2π .

For every nonzero integer n , $F(n) = 0$, so $0 \leq U \leq F$ gives $U(n) = 0$. Because U is real analytic and nonnegative on $\mathbb{R}$, each such zero has even order, hence order at least two. The zeros of

$$F(z) = \left(\frac{\sin \pi z}{\pi z} \right)^2$$

are exactly those nonzero integers, all double. Consequently

$$Q(z) = \frac{U(z)}{F(z)}$$

is entire.

Fix small disjoint disks about the nonzero integers. Outside those disks, the elementary lower bound for the sine gives

$$|F(z)| \geq c \frac{e^{2\pi|\Im z|}}{(1+|z|)^2}, \quad |U(z)| \leq \|u\|_1 e^{2\pi|\Im z|}.$$

Hence $|Q(z)| \leq C(1+|z|)^2$ there. The maximum principle on each omitted disk supplies the same global bound. Therefore Q is a polynomial of degree at most two. On $\mathbb{R}$, however,

$$0 \leq Q(x) = \frac{U(x)}{F(x)} \leq 1$$

away from the zeros, and continuity fills in those points. A polynomial bounded on $\mathbb{R}$ is constant. Write $Q \equiv \lambda$, where $0 \leq \lambda \leq 1$. Then $U = \lambda F$, so $G = \lambda F e^{-\pi\xi^2} = \lambda \widehat{h}$. Fourier injectivity gives $g = \lambda h$, proving extremality. $\square$

Scope and provenance

This is a full affirmative answer to the first clause of Question 3, not a classification of all extreme rays. A separate existing packet in this run proves the second clause false by an explicit perturbative decomposition. Thus the two packets together answer both clauses: $f * \gamma$ is extremal, but $\gamma(f * \gamma)$ is not. Exact arXiv-id, title, quotation, formula, product/convolution, and Borisov-conjecture searches through 13 August 2026 found no later answer to the first clause. The proof above is a new result of this run.

References

- [1] M. Gaál, S. Nagy, and Sz. Gy. Révész, *Integral comparisons of nonnegative positive definite functions on LCA groups*, arXiv:1803.06409 (2018; revised 2019).
- [2] P. Jaming, M. Matolcsi, and Sz. Gy. Révész, *On the extremal rays of the cone of positive, positive definite functions*, arXiv:0801.0941; J. Fourier Anal. Appl. 15 (2009), 561–582.